Instructions for displacement threshold energy exercise

Introduction

To give you a bit of experience with materials simulation we have provided a simple modelling exercise that allows you to calculate the displacement threshold energy in a model material using classical molecular dynamics simulations. These simulations will be run via Google's Colab platform

You can use your university Google account for this.

Setting up Google Colab

The first step is to navigate to the github repository for the session

You should see something like in Figure 1.

Make sure you are viewing `ReadMe.md` and right-click the badge circled in the figure and select `open in new tab`. This will open a Colab tab and you should then see something like Figure 2.

Select `01-DisplacementThreshold/01-DisplacementThreshold.ipynb`. You'll be asked a couple of questions. Confirm you want to proceed you'll then see the notebook in Colab as in Figure 3

If you click the file icon as shown in Figure 3 this will open your folder structure and you should see something like Figure 4.

The next step is to run the two code cells in this notebook. Click on the first code cell and then hit `shift + return`. You'll be asked to sign in to your Google account as in Figure 5.

Of the various permissions, you should only need to select those permissions that are obviously relevant to accessing your files on Google drive. You may need to hit the refresh button in the sidebar (Figure 6) but you should then see that `drive` now appears.

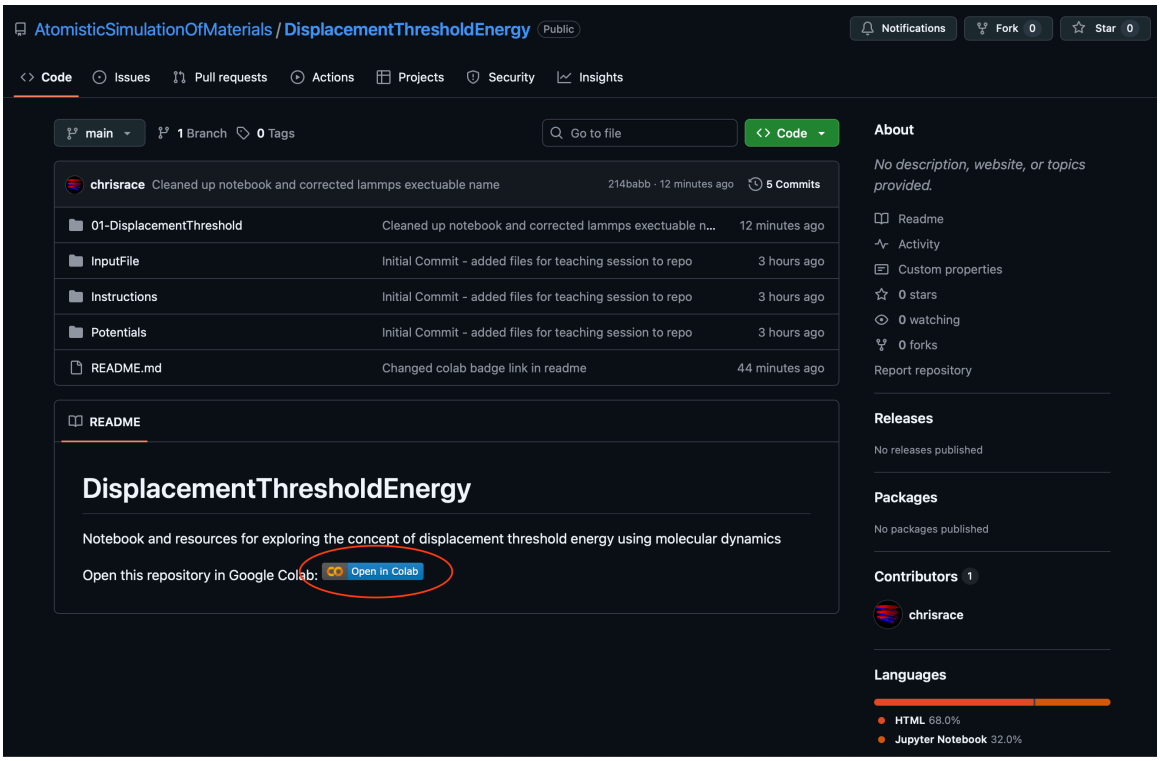

AtomisticSimulationOfMaterials / DisplacementThresholdEnergy
Public
Notifications
Fork 0
Star 0
Code
Issues
Pull requests
Actions
Projects
Security
Insights
main
1 Branch
0 Tags
Go to file
Code
chrisrace Cleaned up notebook and corrected lammps exectuable name
214babb · 12 minutes ago
5 Commits
01-DisplacementThreshold
Cleaned up notebook and corrected lammps exectuable n...
12 minutes ago
InputFile
Initial Commit - added files for teaching session to repo
3 hours ago
Instructions
Initial Commit - added files for teaching session to repo
3 hours ago
Potentials
Initial Commit - added files for teaching session to repo
3 hours ago
README.md
Changed colab badge link in readme
44 minutes ago
README
DisplacementThresholdEnergy
Notebook and resources for exploring the concept of displacement threshold energy using molecular dynamics
Open this repository in Google Colab:
Open in Colab
About
No description, website, or topics provided.
Readme
Activity
Custom properties
0 stars
0 watching
0 forks
Report repository
Releases
No releases published
Packages
No packages published
Contributors 1
chrisrace
Languages
HTML 68.0%
Jupyter Notebook 32.0%

Figure 1

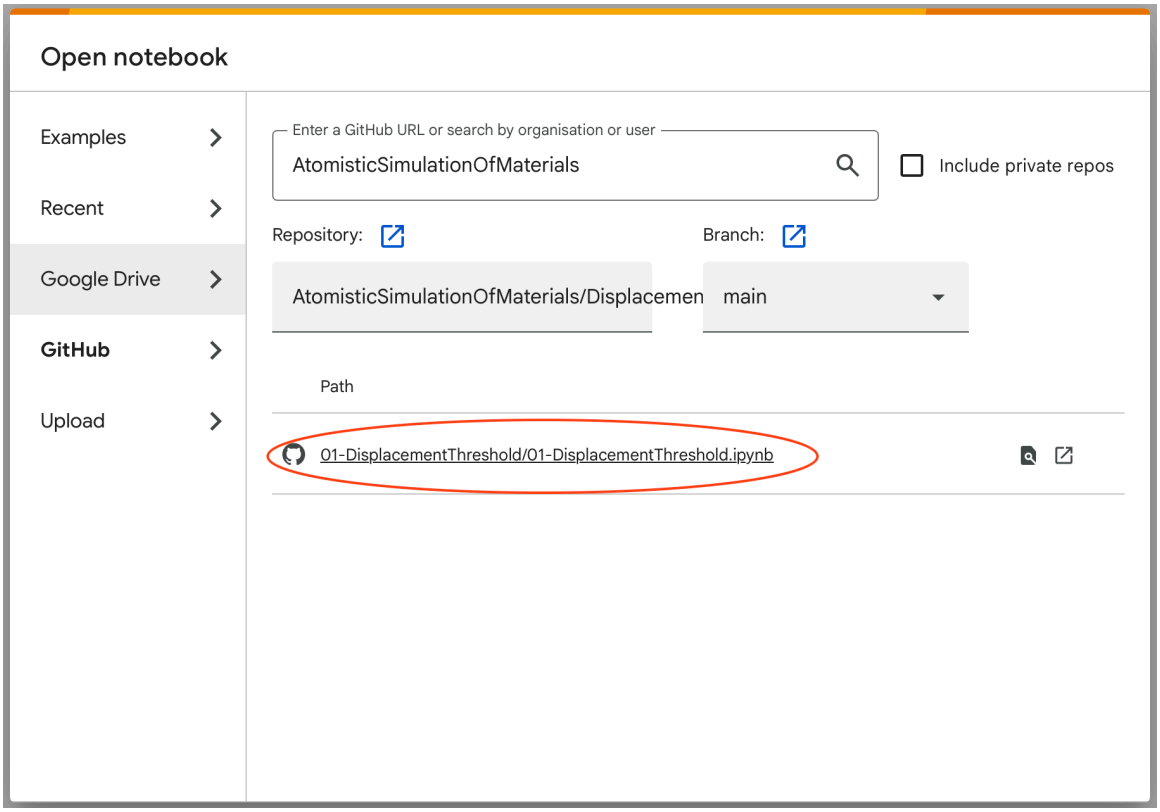

Open notebook
Examples
Recent
Google Drive
GitHub
Upload
Enter a GitHub URL or search by organisation or user
AtomisticSimulationOfMaterials
Include private repos
Repository:
Branch:
AtomisticSimulationOfMaterials/Displacemen
main
Path
01-DisplacementThreshold/01-DisplacementThreshold.ipynb

Figure 2

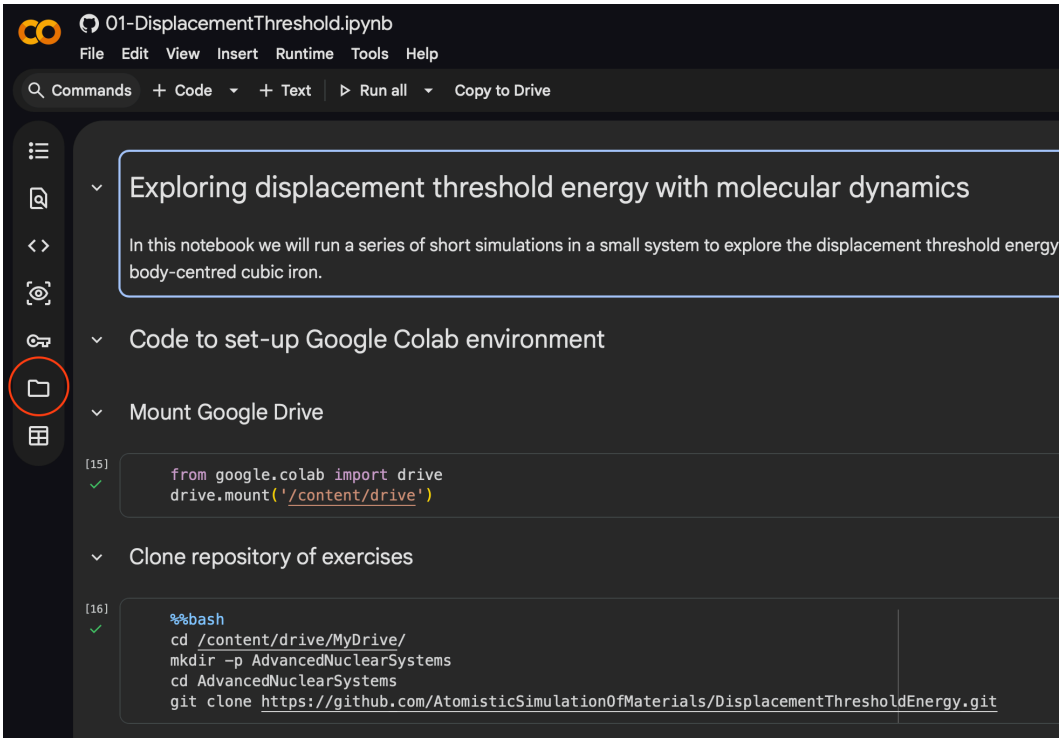

01-DisplacementThreshold.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all Copy to Drive
Exploring displacement threshold energy with molecular dynamics
In this notebook we will run a series of short simulations in a small system to explore the displacement threshold energy
body-centred cubic iron.
Code to set-up Google Colab environment
Mount Google Drive
[15]
from google.colab import drive
drive.mount('/content/drive')
Clone repository of exercises
[16]
%%bash
cd /content/drive/MyDrive/
mkdir -p AdvancedNuclearSystems
cd AdvancedNuclearSystems
git clone https://github.com/AtomisticSimulationOfMaterials/DisplacementThresholdEnergy.git

Figure 3

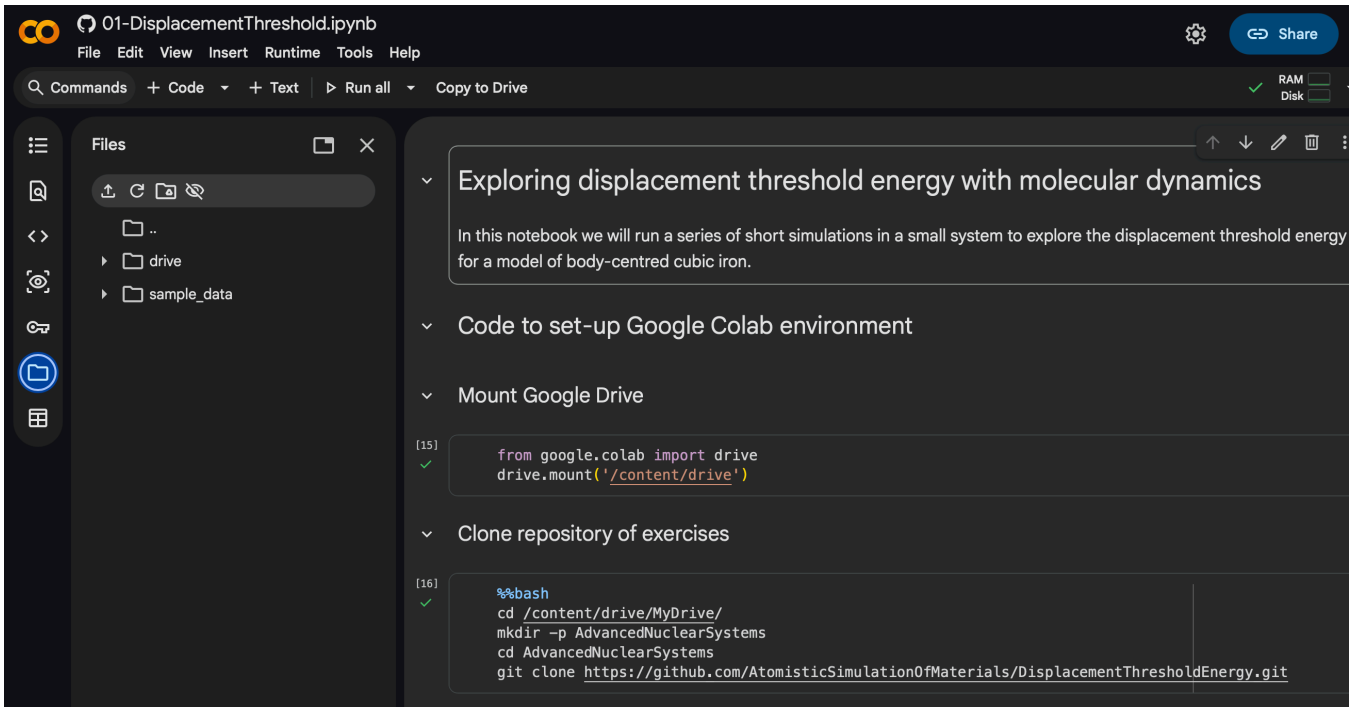


Figure 4

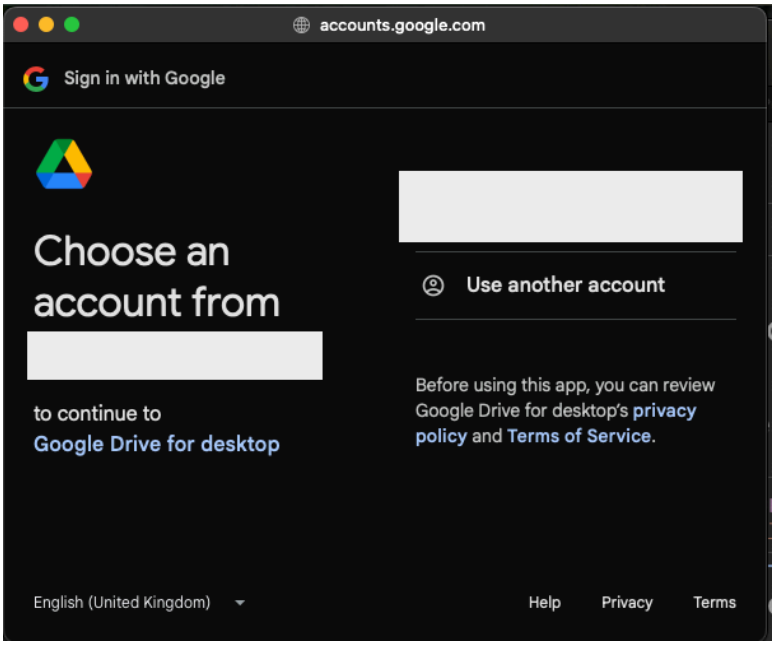


Figure 5

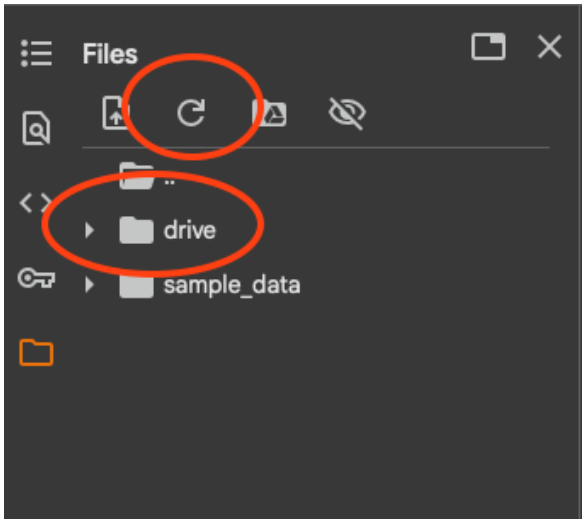


Figure 6

Execute the second code cell in the notebook (by clicking on it then hitting `shift + return`). This will clone the repository of files for the exercises into your Google drive. You should now be able to find all the files for the session under `drive/MyDrive` as in Figure 7 (my Google drive also has a lot of clutter in it).

Run the next three code cells (see Figure 8). These cells:

- Install the lammps molecular dynamics software (and a module that will help us read the output)
- Load the python modules that we will need for the exercises
- Change to the correct directory to run the exercises

The exercises

All that remains now is to work through the cells in the rest of the notebook.

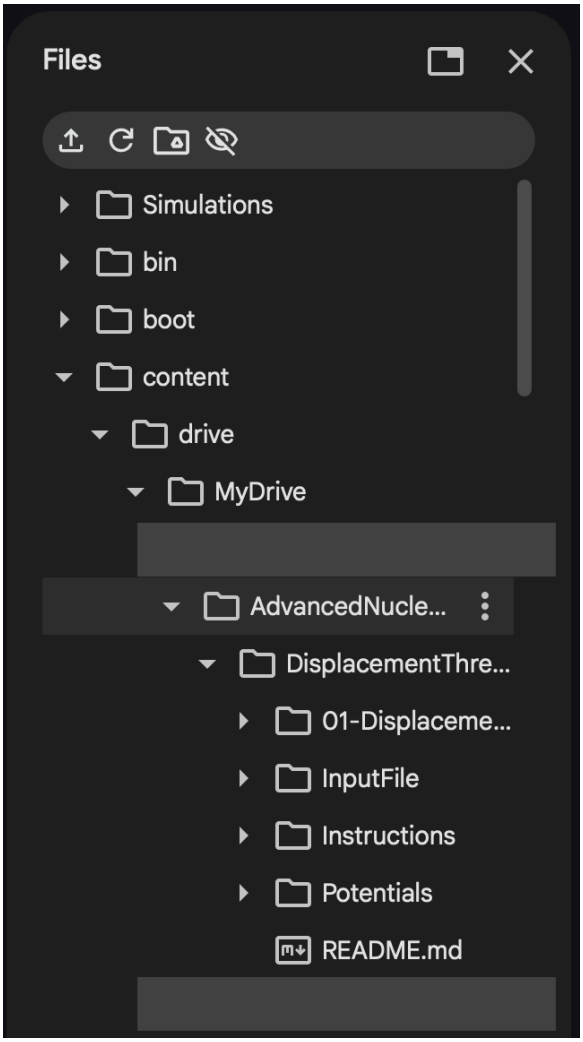
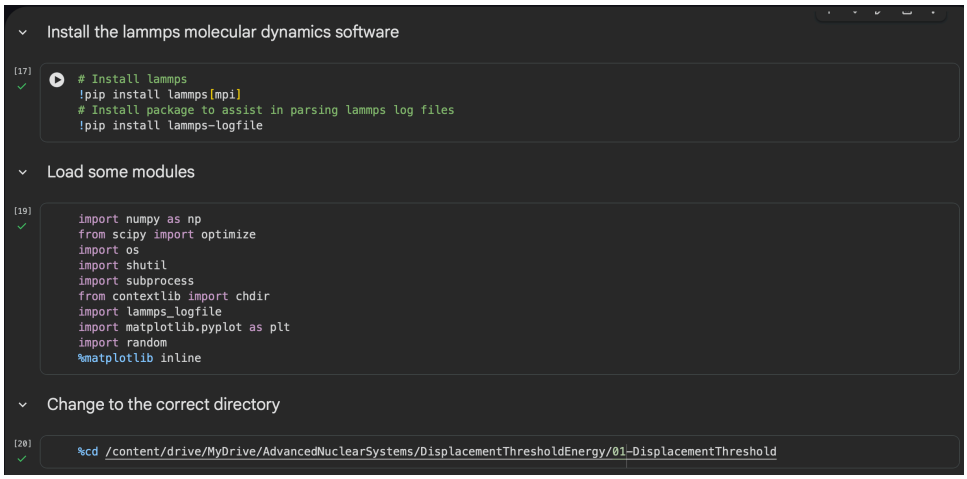

Files
Simulations
bin
boot
content
drive
MyDrive
AdvancedNucle...
DisplacementThre...
01-Displaceme...
InputFile
Instructions
Potentials
README.md

Figure 7



Install the lammps molecular dynamics software

[17]

```
# Install lammps
!pip install lammps[mpi]
# Install package to assist in parsing lammps log files
!pip install lammps-logfile
```

Load some modules

[19]

```
import numpy as np
from scipy import optimize
import os
import shutil
import subprocess
from contextlib import chdir
import lammps_logfile
import matplotlib.pyplot as plt
import random
%matplotlib inline
```

Change to the correct directory

[20]

```
%cd /content/drive/MyDrive/AdvancedNuclearSystems/DisplacementThresholdEnergy/01-DisplacementThreshold
```

Figure 8